

Nurse Lab Reports

Frontend Developer Guide

Version: 1.0 | Date: 24 August 2026 | Audience: Frontend developers

Backend: HM-System-Backend (FastAPI) | Auth: JWT Bearer token

Nurse prefix: /nurse/lab-reports | Read-only (view + download)

BACKEND IS DONE - FRONTEND IS NOT WIRED

Use this guide to add a Nurse Lab Reports screen the same way Vitals and Nursing Notes work. Do NOT call lab technician APIs (/lab/reports). Do NOT add upload, edit, complete, or delete actions for nurses.

1. Purpose

Nurses must be able to view and download completed lab reports for patients who currently occupy a hospital bed. This matches the nurse clinical workflow (patient name + ward + bed + doctor), not the Lab Technician work queue.

- VIEW list and report detail: YES
- DOWNLOAD report file (PDF/image): YES
- UPLOAD / UPDATE / DELETE / COMPLETE: NO
- No new database tables. Reuses lab_test_orders + lab_results
- No Alembic migration. After pull, run: python seed.py

2. Critical: do not reuse Lab Technician APIs

Wrong (lab tech)	Correct (nurse)
GET /lab/reports	GET /nurse/lab-reports
GET /lab/reports/{id}	GET /nurse/lab-reports/{report_id}
GET /lab/reports/{id}/file	GET /nurse/lab-reports/{report_id}/file

GET /lab/reports is filtered by the lab technician department (Laboratory vs Radiology). A nurse token will 403 or see the wrong data. Do not reuse useLabReportsQuery, lab.js, or LabCompletedReportsPage.

3. Permission (one key only)

Permission	Role	Used for
nurse_lab_reports:view	nurse	List, detail, and file download

All requests require: Authorization: Bearer <access_token>

401 = token missing/expired. 403 = permission missing.

There is NO nurse_lab_reports:create / update / delete.

- Add key to useNursePermission.js (labReportsView)
- Add key to seedPermissions.js and adminEditLocks.js nurse_clinical
- Add a View-only group in nurseManagementConfig.js (Lab Reports)
- Hide the nav item when the nurse does not have this permission

4. API endpoints (3 total)

#	Method	URL	Purpose
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1	GET	/nurse/lab-reports	Paginated report list
2	GET	/nurse/lab-reports/{report_id}	Report detail + parameters
3	GET	/nurse/lab-reports/{report_id}/file	Download file (blob)

No POST / PUT / PATCH / DELETE under /nurse/lab-reports.

5. Patient scope (same All / Allocated bar as Vitals)

Reuse NursePatientScopeContext. Pass ...scopeFilters on EVERY call (list, detail, and file). Detail and file also honor allocated_only. If you omit it on detail/file, the nurse can open a report that is not in the current Allocated list.

Mode	Query	Who the nurse sees
All (default)	allocated_only omitted or false	Reports for patients currently occupying ANY hospital bed
Allocated	allocated_only=true	Reports for patients on THIS nurse's allocated occupied beds

NOT THE SAME AS LAB TECH OR OPD LAB HISTORY

OPD patients with no occupied bed are excluded. Discharged patients drop out of the list when the bed is freed. This is intentional: nurses see reports for patients currently assigned to beds.

Optional with allocated_only=true (same as vitals/notes):

- assignment_date (defaults to today IST)
- shift_name (defaults to current duty shift)

6. GET /nurse/lab-reports List

Query parameters

Param	Type	Default	Description
search	string	-	Patient name, UHID, or test name
patient_id	int	-	Internal patient id (must be in scope)
patient_uid	string	-	UHID e.g. P-1037
patient_name	string	-	Name contains
test_name	string	-	Test name contains
from_date	date	-	YYYY-MM-DD (report created_at)
to_date	date	-	YYYY-MM-DD inclusive
allocated_only	bool	false	Allocated beds only
assignment_date	date	-	With allocated_only
shift_name	string	-	With allocated_only
page	int	1	Page number
page_size	int	20	Max 100

Example request

```
GET /nurse/lab-reports?search=CBC&allocated_only=true&page=1&page_size=20
Authorization: Bearer <nurse_token>
```

Success response (200)

```
{
  "success": true,
  "total": 2,
  "page": 1,
  "page_size": 20,
  "items": [
    {
      "report_id": 41,
      "order_id": 88,
      "patient_id": 1037,
      "patient_name": "Sumit Sharma",
      "patient_uid": "P-1037",
      "registration_source": "IPD",
      "encounter_type": "IPD",
      "admission_id": 12,
      "ward_name": "General",
      "bed_number": "G-12",
      "doctor_id": 9,
      "doctor_name": "Dr. Arora",
      "test_name": "CBC",
      "uploaded_by": 22,
      "uploaded_by_name": "Lab Tech One",
      "report_file": "uploads/lab_reports/....pdf",
      "uploaded_at": "2026-08-24T10:15:00+05:30",
      "status": "completed",
      "source": "BOTH"
    }
  ]
}
```

}

List item fields to display

API field	UI label	Notes
patient_uid	Patient ID	Show UHID, never internal patient_id
patient_name	Patient Name	Required column
ward_name	Ward	From occupied bed; may be null
bed_number	Bed	From occupied bed; may be null
doctor_name	Doctor	Attending / ordering doctor with Dr. prefix
test_name	Test	e.g. CBC, X-Ray Chest
source	Type	PARAMETERS PDF BOTH NONE
uploaded_at	Reported at	ISO datetime IST
status	Status	Usually completed
report_file	Has file	Non-null => show Download

source meaning: PARAMETERS = result values only, PDF = file only, BOTH = values + file, NONE = empty (rare; hide download).

7. GET /nurse/lab-reports/{report_id} Detail

Use for a detail page or modal. Pass the same allocated_only / assignment_date / shift_name as the list that opened this report.

Path

- report_id = items[].report_id from the list (integer, not RPT-41)

Query (same scope as list)

Param	Type	Default
allocated_only	bool	false
assignment_date	date	today IST if allocated
shift_name	string	current shift if allocated

Success response (200) note id vs list report_id

List uses report_id. Detail uses id for the same row. Map both: reportDbId = row.report_id ?? row.id

```
{
  "id": 41,
  "lab_test_order_id": 88,
  "uploaded_by": 22,
  "uploaded_by_name": "Lab Tech One",
  "sample_collected_at": "2026-08-24T09:00:00+05:30",
  "test_performed_at": "2026-08-24T10:00:00+05:30",
  "report_file": "uploads/lab_reports/....pdf",
  "remarks": "Within normal limits",
  "created_at": "2026-08-24T10:15:00+05:30",
  "file_name": "cbc-p1037.pdf",
  "file_type": "application/pdf",
  "file_size": 184320,
  "file_size_display": "180.0 KB",
  "source": "BOTH",
  "order": {
    "id": 88,
    "patient_id": 1037,
    "patient_name": "Sumit Sharma",
    "patient_uid": "P-1037",
    "registration_source": "IPD",
    "encounter_type": "IPD",
    "admission_id": 12,
    "ward_name": "General",
    "bed_number": "G-12",
    "doctor_id": 9,
    "doctor_name": "Dr. Arora",
    "department_id": 11,
    "test_name": "CBC",
    "category": "Hematology",
    "priority": "Normal",
    "status": "completed"
  },
  "parameters": [
    {
      "id": 1,
      "parameter_name": "Hemoglobin",
```

```

    "value": "13.2",
    "unit": "g/dL",
    "normal_range": "12-16",
    "flag": "normal"
  }
]
}

```

parameter.flag is normal | low | high | null. Render a tone badge. If parameters is [] and report_file is set, show Download only.

8. GET /nurse/lab-reports/{report_id}/file Download

This is NOT JSON. Response is a file stream (FileResponse). Copy the lab technician blob pattern, but change the URL.

```

const response = await fetch(
  `${API_BASE_URL}${API_PREFIX}/nurse/lab-reports/${reportId}/file?allocated_only=true`,
  { headers: { Authorization: `Bearer ${token}` } }
);
const blob = await response.blob();
const fileName = parseContentDisposition(response) || `lab-report-${reportId}`;
// then URL.createObjectURL(blob) + <a download>

```

- Pass the same allocated_only query string as the list
- 404 if no file uploaded, file missing on disk, or patient out of scope
- Content-Type comes from the server (application/pdf or image/*)
- Do not use apiClient JSON helper for this call

9. Errors

Status	When	UI action
401	Missing/expired JWT	Redirect to login
403	Missing nurse_lab_reports:view	Hide nav and page
404	Unknown report_id OR patient not on an allowed occupied bed	Show not found; do not leak why
404	File endpoint: no report_file or file gone from disk	Hide Download / toast 'No file'
200 + empty items	No occupied beds, or no allocated beds this shift	Empty state, not an error

If allocated_only=true and the nurse has no bed allocations, total=0 and items=[]. Reuse the vitals empty copy: 'No beds assigned for this shift.'

10. Suggested screens (mirror Vitals / Notes)

Screen	Route	APIs
Lab Reports registry	/nurse/lab-reports	GET /nurse/lab-reports
Report detail	/nurse/lab-reports/:reportId	GET /{id} + optional file
Patient overview tab	/nurse/patients/:patientId	GET list?patient_id=

Registry table columns (same pattern as NurseVitalsRegistryPage)

- Patient ID (UHID) | Patient Name | Ward | Bed | Doctor | Test | Type | Reported at | Actions
- Actions: View (always) + Download (only if report_file is set)
- Row click can open detail. No Edit / Upload buttons.
- Search box: send as search= (backend matches name, UHID, test name)
- Pagination: page + page_size; use total from the payload
- Wire NursePatientScopeBar: ...scopeFilters on the query

Patient overview tab

On NursePatientOverviewPage add a Lab Reports tab next to Vitals / Notes / Meds. Call GET /nurse/lab-reports?patient_id={id}&page_size=20 plus scopeFilters. If the patient is not currently on an occupied bed, the list is empty.

Nav

Add 'Lab Reports' to NurseLayout after Nursing Notes (or after Medications). Show only when canViewLabReports is true. Suggested icon: FlaskConical or FileSearch from lucide-react.

11. Frontend files to add or change

Do not modify lab technician feature files. Touch nurse + shared nurse API only.

```

ADD
src/features/nurse/pages/NurseLabReportsRegistryPage.jsx
src/features/nurse/pages/NurseLabReportDetailPage.jsx    (or a modal)

CHANGE
src/features/nurse/api/nurse.js
  getLabReports(params, token)
  getLabReportById(reportId, params, token)

```

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```
fetchNurseLabReportFileBlob(reportId, params, token) // raw fetch, not JSON

src/shared/hooks/queries/useNurseQuery.js
  useNurseLabReportsQuery(filters)
  useNurseLabReportQuery(reportId, filters)

src/shared/api/queryKeys.js
  nurse.labReports(filters)
  nurse.labReport(id, filters)

src/shared/api/mappers/nurseMapper.js
  mapLabReportItem(row) // report_id, patient_uid, ward, bed, doctor_name
  // You MAY reuse mapApiVisitLocationFields from labMapper.js
  // Do NOT import lab services / lab query hooks

src/features/nurse/hooks/useNursePermission.js
  labReportsView: 'nurse_lab_reports:view'

src/features/nurse/components/NurseLayout.jsx
  nav link Lab Reports

src/routes/nurseRoutes.jsx
src/shared/constants/index.js
  NURSE_LAB_REPORTS, NURSE_LAB_REPORT_DETAIL

src/features/admin/nurse/constants/nurseManagementConfig.js
src/features/admin/constants/adminEditLocks.js
src/features/super-admin/constants/seedPermissions.js
  nurse_lab_reports:view (view only)

OPTIONAL
src/features/nurse/pages/NursePatientOverviewPage.jsx
  Lab Reports tab
```


12. Suggested service wrappers

```
// nurse.js
export function getNurseLabReports(params, token) {
  return apiClient(appendQuery('/nurse/lab-reports', params), { token });
}

export function getNurseLabReportById(reportId, params, token) {
  return apiClient(
    appendQuery(`/nurse/lab-reports/${reportId}`, params),
    { token }
  );
}

// file download: copy fetchLabReportFileBlob but URL =
// /nurse/lab-reports/${reportId}/file + same query params
```

Always forward `allocated_only` from `scopeFilters`. Example list call:

```
useNurseLabReportsQuery({
  search: debouncedSearch || undefined,
  page,
  page_size: 20,
  ...scopeFilters, // { allocated_only: true } when Allocated mode
})
```

13. Display rules

- Patient ID column = `patient_uid` (P-1037). Never show numeric `patient_id`.
- Doctor column = `doctor_name` (already includes Dr. prefix). Fallback em dash.
- Ward / Bed = `ward_name` / `bed_number`. Same cells as dashboard / vitals.
- Empty ward/bed: show em dash. Do not invent values on the client.
- Read-only: no forms, no file input, no status change buttons.
- Print is optional; if you print, use detail JSON, do not call `/lab print` helpers that hit `/lab/reports`.

14. What backend already does (do not reimplement)

- Occupied-bed filter (All) and allocated-bed filter (Allocated)
- 404 for out-of-scope report ids (treat as not found)
- Doctor name snapshot with Dr. prefix
- Ward/bed from the occupied bed row
- Audit log on detail view and file download (Super Admin audit page)
- Pagination total / page / page_size / items

15. Backend setup (local / devops)

```
cd HM-System-Backend
python seed.py
# Adds nurse_lab_reports:view to the nurse role.
# No alembic migration for this feature.
```

Until `seed.py` is run, nurse calls return 403.

Aligned with `Routers/nurse_lab_report_router.py` and `Services/nurse_lab_report_service.py` (24 August 2026).

16. QA checklist for frontend

- Nurse without permission: nav hidden, direct URL blocked
- All mode: sees reports for any currently occupied bed
- Allocated mode: only allocated occupied beds; empty if none assigned
- Search by name / UHID / test name updates the list
- View opens detail with parameters table
- Download works for PDF and is hidden when report_file is null
- OPD-only patient (no bed) never appears
- Lab technician pages still use /lab/reports unchanged
- No upload/edit UI anywhere in nurse module